

```
// is the simplest possible example, but we'll cover more
advanced
// cases with branching conversations, conditional next
states, memory,
// jumps and more in other guides.
//
// For more information on state machines in general, see:
// https://en.wikipedia.org/wiki/Finite-state_machine
sm = dt.NewStateMachine(p)
sm.SetStates(
  []dt.State{
    {
      OnEntry: func(in *dt.Msg) string {
        return "Hello world!"
      },
      OnInput: func(in *dt.Msg) {
      },
      Complete: func(in *dt.Msg) (bool, string) {
        return true, ""
      },
    },
  },
)

// Run is called when the user first enters the package in a
conversation. Here
// we simply reset the state of the state machine and pass it
on to FollowUp().
func Run(in *dt.Msg) (string, error) {
  sm.Reset(in)
  return FollowUp(in, resp)
}

// FollowUp is called as the user continues to message the
package after the
// user's first message. Here we tell the state machine to
move to the next
// available state and return the response from that move--
since there's only
// one state, it will always be "Hello world!" from the
OnEntry function.
func FollowUp(in *dt.Msg) (string, error) {
  return sm.Next(in), nil
}

You can find and edit this file at $GOPATH/src/github.com/itsabot/plugin_hello. You should see Abot boot with a line or two mentioning our new plugin, 'Hello World!'. Open another terminal while the Abot server is still running, and type (using the same phone number you used to sign up at http://localhost:4200):
```

```
$ abot console +918889992221
> Say hi
Hello World!
```

Abot just routed your message to the plugin based on the trigger defined in our `abot_hello.go`. The state machine told it to respond with 'Hello World!' when it entered its first state, and since there are no other states, this state is replayed every time a new message matching our trigger is sent to Abot.

Deploying Abot

To make Abot available to the outside world, we'll have to deploy it to a server. We will be using Heroku for now. Go and Abot do make it easy to work on any Web server, though. Open a terminal and run the following code:

```
$ echo 'web: abot server' > Procfile
$ heroku create --buildpack https://github.com/itsabot/
heroku-buildpack-go
$ heroku addons:create heroku-postgresql:hobby-dev --version
9.5
$ cat db/migrations/up/*.sql | heroku pg:psql
$ cat data/cities.csv | heroku pg:psql -c "COPY cities(name,
countrycode) FROM stdin DELIMITER ',' CSV;"
$ heroku config:add ABOT_ENV=production \
ABOT_URL="https://YOURURL" \
ABOT_SECRET=$(cat /dev/urandom tr -dc _A-Z-a-z-0-9 | head
-c${1:-64};echo;)
$ git push heroku master
$ heroku open
```

Be sure to replace YOURURL above with a user-presentable URL. If everything booted correctly, you'll see the 'Congratulations, you're running Abot!' screen. If not, you can track down any issues with `heroku logs --tail`.

Once you see the 'Congratulations' screen, try communicating with your Abot. Sign up for an account using the Web interface, then use the following commands:

```
$ abot console https://yourapp.herokuapp.com +yourphone
> Hi
Hello there!
```

Replace *yourapp* and *yourphone* with the appropriate values, and you should see Abot respond to your message!

Abot is a new framework, and more versions will be released soon. One can also contribute to the project. 

References

- [1] <https://github.com/itsabot/abot>
- [2] [https://en.wikipedia.org/wiki/Go_\(programming_language\)](https://en.wikipedia.org/wiki/Go_(programming_language))
- [3] <https://github.com/itsabot/abot/wiki/Roadmap>

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